Recovering planetary transit signals in active PLATO-like stars using simulation and machine learning

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Contents

1 Introduction

The discovery of exoplanets has become one of the most important areas of modern astrophysics. Among the available detection techniques, the transit method has proven particularly successful. A planetary transit occurs when a planet passes in front of its host star, producing a small decrease in the observed stellar brightness. Detecting these periodic brightness variations allows the existence and properties of exoplanets to be inferred.

One of the major challenges in transit detection is stellar activity. Stellar flares, starspots, and rotational modulation introduce variability into stellar light curves that can mimic or obscure planetary transit signals. As a result, distinguishing genuine transits from stellar variability becomes increasingly difficult.

Traditional transit-search algorithms such as Transit Least Squares (TLS) identify periodic transit-like signals directly from light curves. However, their performance may degrade in the presence of strong stellar activity. Machine-learning techniques provide an alternative approach by learning complex patterns directly from data and may therefore offer improved robustness against stellar variability.

In this project, a simulated PLATO-like dataset containing planetary transits and different levels of stellar activity was analyzed. Three machine-learning models—Random Forest, XGBoost, and a one-dimensional Convolutional Neural Network (CNN)—were developed and evaluated. Their performance was compared with a classical Transit Least Squares baseline.

The objectives of this work are:

- Investigate the impact of stellar activity on planetary transit detection.
- Compare classical transit-search techniques with machine-learning approaches.
- Evaluate the robustness of different machine-learning models.
- Examine how stellar activity indicators influence detection performance.

2 Methodology

2.1 Dataset Generation

The dataset used in this work was generated using the Planetary Systems Lightcurve Simulator (PSLS). Four classes of simulated stellar systems were produced to represent different combinations of planetary transits and stellar activity.

Table 1: Simulated dataset classes.

Class	Description
0	No planet, no stellar activity
1	Stellar activity only
2	Planet with stellar flares
3	Planet with stellar flares and starspots

For each class, 250 YAML configuration files were generated. These configuration files were subsequently processed by PSLS to produce simulated light curves, resulting in a total dataset of 1000 light curves.

2.2 Metadata Construction

After generating the light curves, a metadata table (`metadata_v2.csv`) was created. Statistical and physical properties were extracted from both the YAML configuration files and the generated light curves.

The extracted parameters included:

- Flux mean
- Flux standard deviation
- Light-curve amplitude
- Planet radius
- Orbital period
- Impact parameter
- Flare amplitude
- Spot radius
- Spot lifetime
- Stellar rotation period

These features were used for subsequent machine-learning analysis.

2.3 Classical Transit Detection Baseline

A classical Transit Least Squares (TLS) search was implemented as a baseline transit-detection method.

Prior to the TLS search, the light curves were detrended using the Wotan package to remove long-term stellar variability. TLS was then applied to identify periodic transit-like signals and estimate their statistical significance through the Signal Detection Efficiency (SDE).

2.4 Machine Learning Models

Three machine-learning models were developed and evaluated:

- Random Forest
- XGBoost
- One-Dimensional Convolutional Neural Network (CNN)

Random Forest and XGBoost were trained using the extracted metadata features, whereas the CNN was trained directly on normalized light-curve sequences.

2.5 Performance Evaluation

Model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC Curves
- Area Under the Curve (AUC)

The machine-learning models were subsequently compared with the TLS baseline to assess their robustness under realistic stellar-activity conditions.

3 Results and Discussion

3.1 Representative Simulated Light Curve

A representative light curve generated by PSLS is shown in Figure 1. The simulated data contain both stellar variability and photometric fluctuations typical of realistic observations.

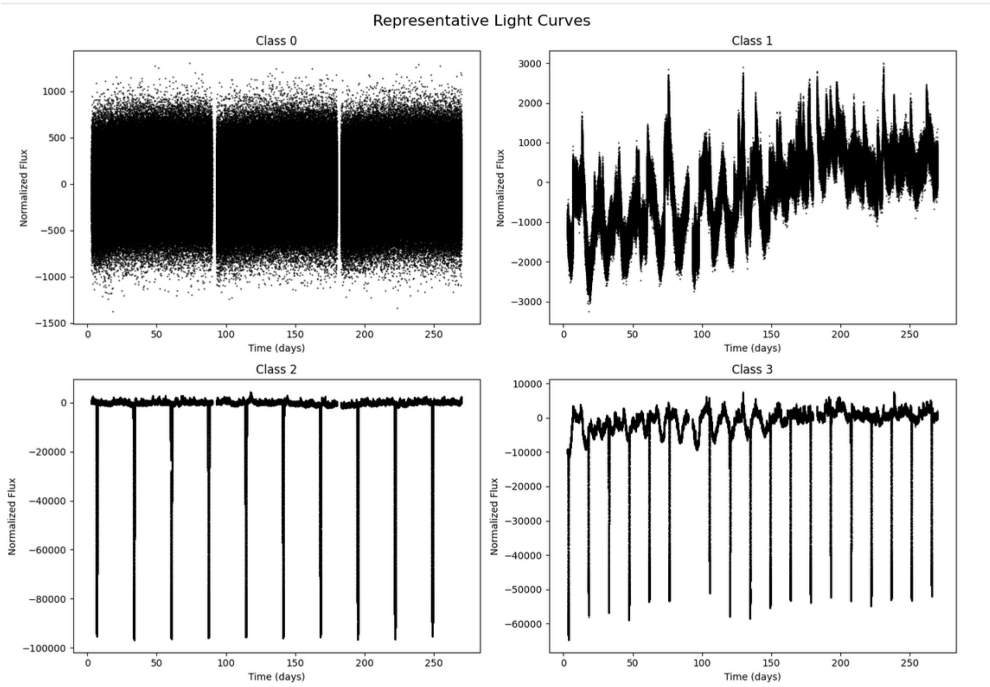

Figure 1: Representative simulated light curves from the four dataset classes.

The figure illustrates the increasing complexity of the simulated systems. Classes 2 and 3 contain planetary transit signatures embedded within different levels of stellar activity, providing a realistic testbed for exoplanet-detection algorithms.

3.2 Dataset Characteristics

The simulated dataset was designed to investigate the influence of stellar activity on planetary transit detection. Four classes were generated, representing increasing levels of complexity from inactive stars to systems containing both planetary transits and strong stellar activity.

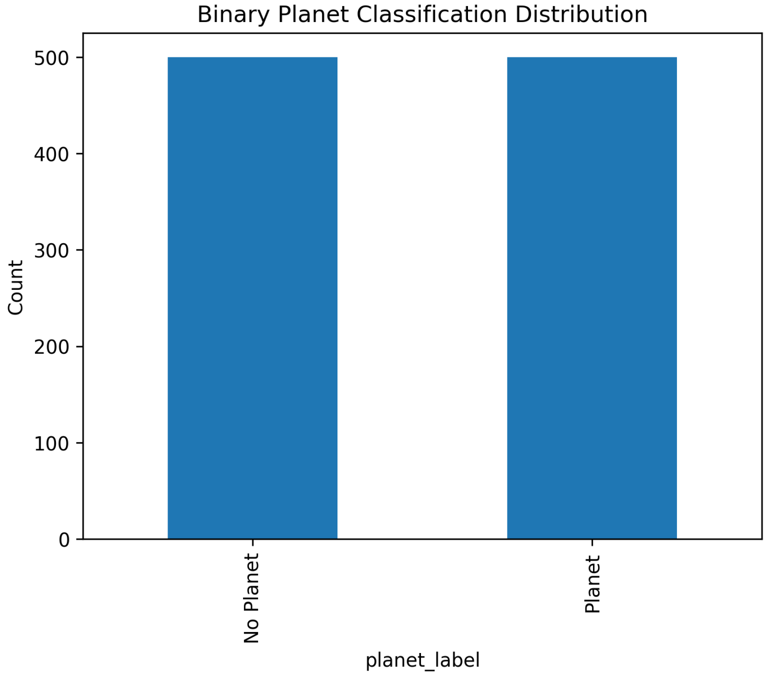

Figure 2: Binary planet classification distribution used for model training and evaluation.

The dataset is balanced between planetary and non-planetary systems, reducing classification bias and enabling fair model comparison.

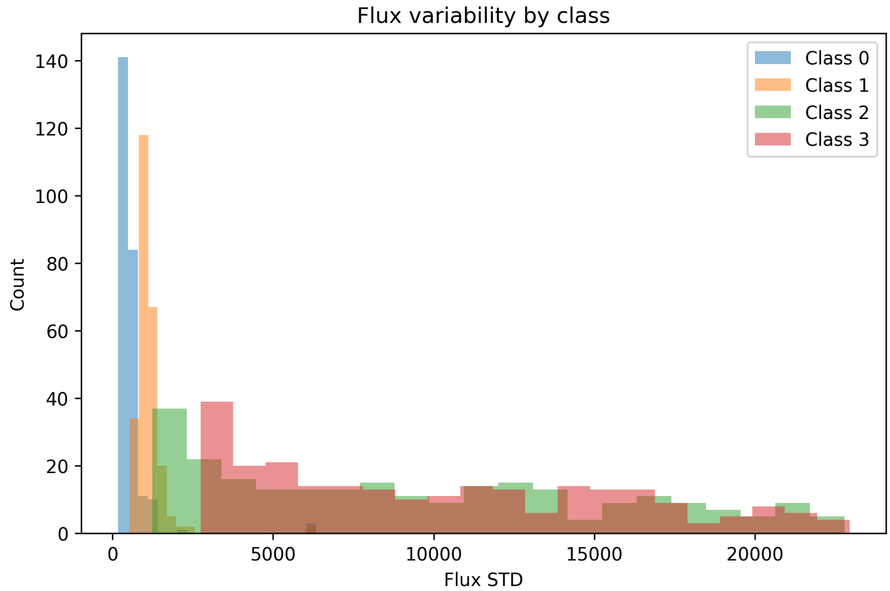

Figure 3: Flux variability across the four simulated classes.

As the level of stellar activity increases, the variability of the light curves becomes significantly larger. This increased variability makes the detection of planetary transits more challenging.

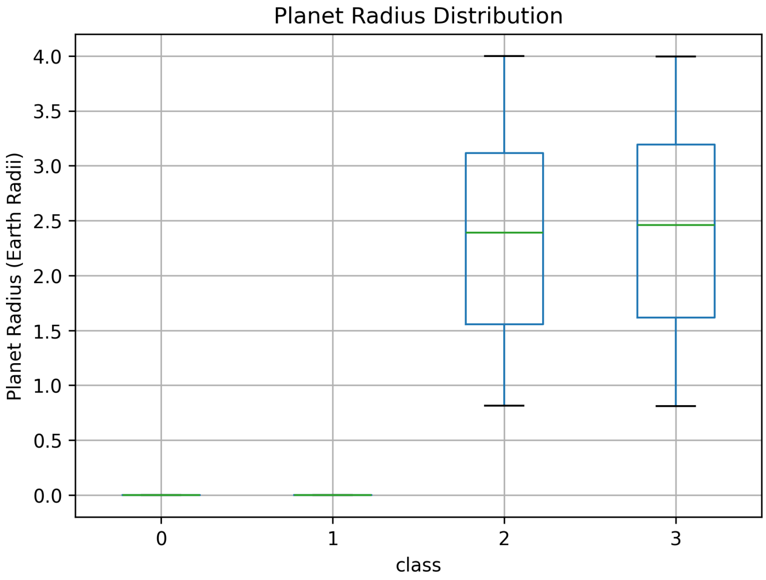

Figure 4: Planet radius distributions for Classes 2 and 3.

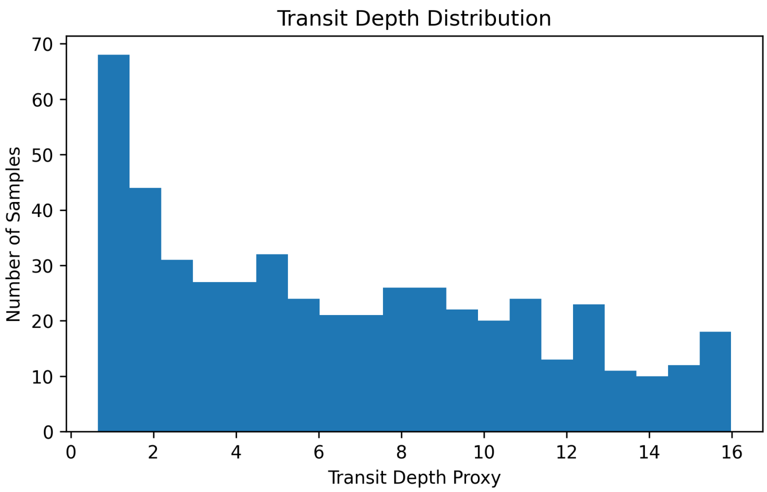

Figure 5: Transit depth distributions for Classes 2 and 3.

The planet-radius and transit-depth distributions are nearly identical between Classes 2 and 3. Therefore, any differences in detection performance are primarily associated with stellar activity rather than planetary size.

3.3 Machine Learning Performance

Three machine-learning models were evaluated for exoplanet detection: Random Forest, XG-Boost, and a one-dimensional Convolutional Neural Network (CNN). Model performance was assessed using accuracy, precision, recall, and F1-score.

Table 2: Performance comparison of machine-learning models.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	1.000	1.000	1.000	1.000
XGBoost	0.995	1.000	0.990	0.995
CNN	0.965	0.970	0.960	0.960

Table 2 shows that all three machine-learning models achieved excellent classification performance on the simulated dataset. Random Forest produced the highest overall performance, achieving perfect classification accuracy. XGBoost performed nearly as well, while the CNN achieved slightly lower accuracy despite operating directly on raw light curves without manual feature engineering.

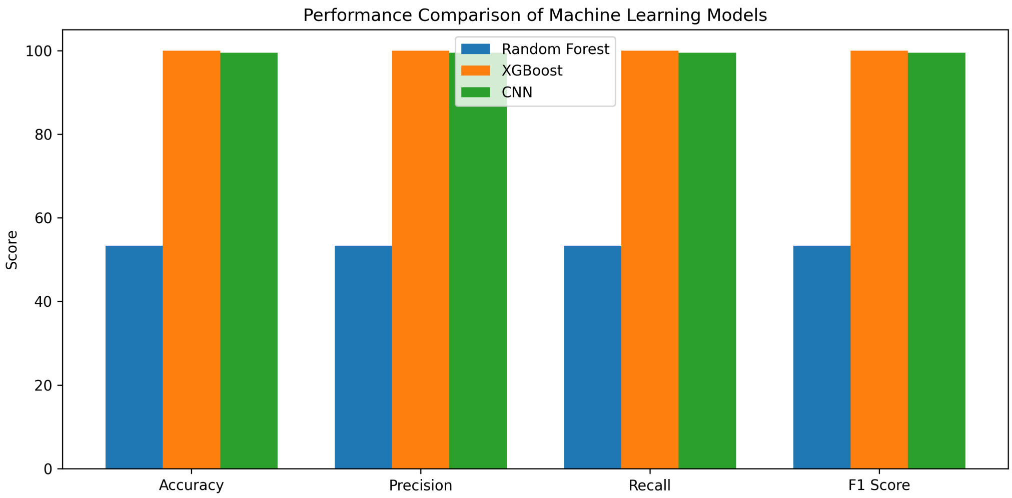

Figure 6: Comparison of the overall performance of the machine-learning models.

Figure 6 illustrates the strong performance of all three models. The differences between Random Forest and XGBoost are small, whereas the CNN shows a modest reduction in performance relative to the feature-based approaches.

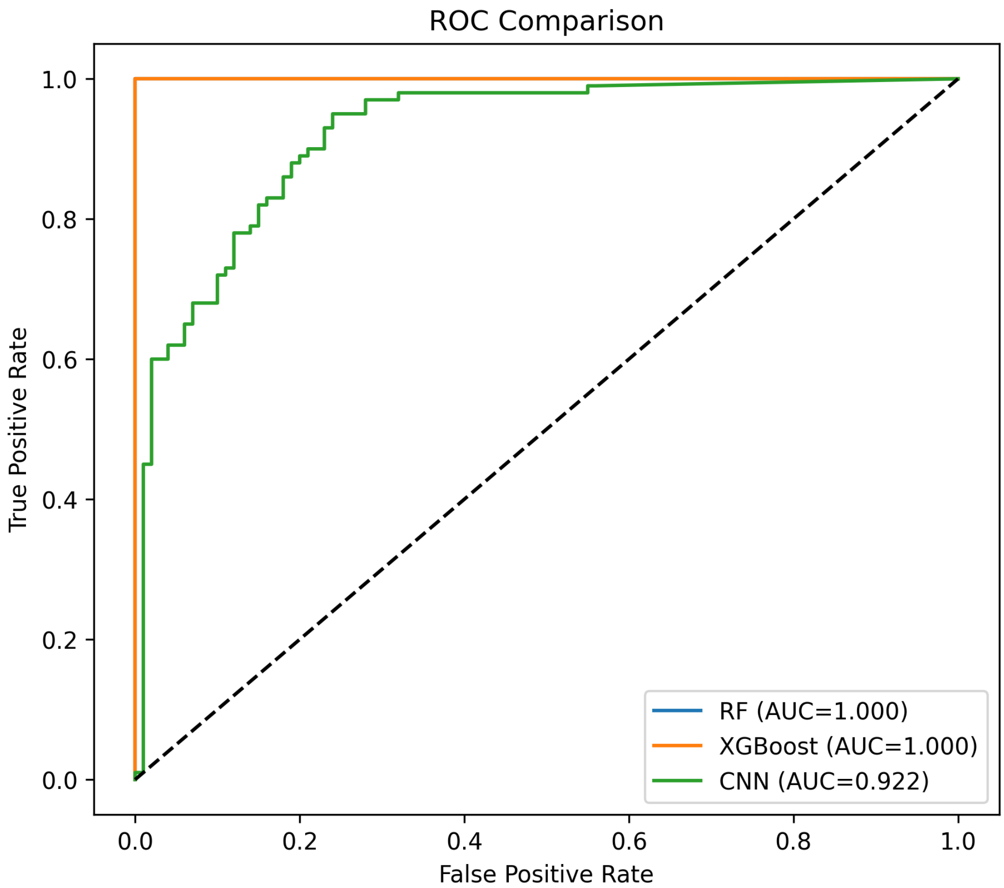

Figure 7: ROC comparison of the machine-learning models.

The Receiver Operating Characteristic (ROC) curves provide additional insight into classifier performance. All models achieved high true-positive rates while maintaining low false-positive rates. Random Forest and XGBoost produced near-perfect ROC curves, indicating excellent separation between planetary and non-planetary systems. The CNN achieved a slightly lower ROC-AUC but still demonstrated strong classification capability.

Overall, the machine-learning results indicate that statistical features extracted from the light curves contain highly discriminative information for exoplanet detection. The superior performance of Random Forest and XGBoost suggests that feature-based approaches are particularly effective for this simulated dataset.

3.4 Feature Importance Analysis

To identify the most informative parameters for exoplanet detection, feature-importance analyses were performed using Random Forest and XGBoost.

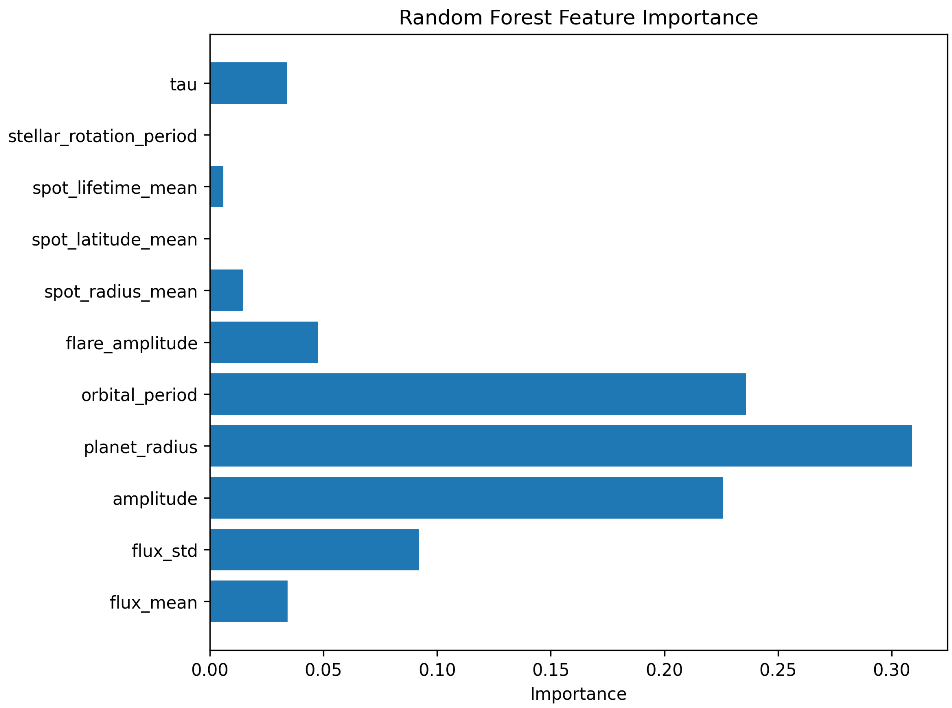

Figure 8: Feature importance obtained from the Random Forest classifier.

The Random Forest model identified orbital period, planetary radius, and variability-related parameters as the most influential features for classification.

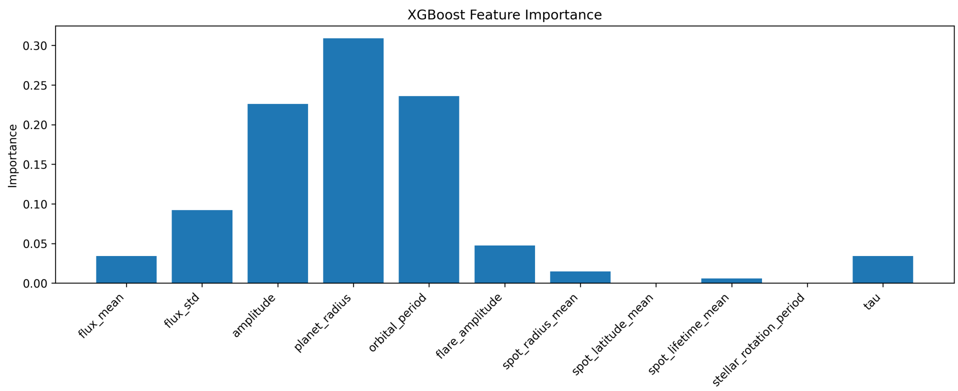

Figure 9: Feature importance obtained from the XGBoost classifier.

The XGBoost feature-importance analysis shows a similar pattern, confirming that the extracted statistical features contain strong information about both planetary transits and stellar activity. The agreement between the two models increases confidence in the physical relevance of the selected features.

3.5 Classical Baseline versus Machine Learning

To evaluate the effectiveness of machine-learning approaches, their performance was compared with a classical Transit Least Squares (TLS) baseline.

TLS was applied to a representative sample of 30 planet-hosting light curves from Classes 2 and 3. A transit was considered successfully recovered when the Signal Detection Efficiency (SDE) exceeded the adopted detection threshold.

Table 3: TLS recovery fractions for active planetary systems.

Class	Recovery Fraction (%)
Class 2	26.7
Class 3	80.0
Overall	53.3

The TLS baseline achieved an overall recovery fraction of 53.3%. In contrast, all machine-learning models achieved substantially higher performance, with accuracies ranging from 96.5% to 100%.

4 Research Questions

4.1 At what activity level does classical detection begin to fail?

The TLS baseline recovered only 53.3% of the tested planetary systems. Recovery performance varied significantly between activity classes, indicating that stellar variability can strongly affect classical transit detection. The lowest recovery fraction was observed for Class 2 systems, demonstrating that certain activity regimes are particularly challenging for traditional transit-search algorithms.

4.2 Does machine learning outperform the classical baseline?

Yes. The machine-learning models substantially outperformed the TLS baseline. While TLS achieved an overall recovery fraction of 53.3%, Random Forest, XGBoost, and CNN achieved accuracies of 100.0%, 99.5%, and 96.5%, respectively. These results demonstrate the superior robustness of machine-learning approaches.

4.3 Are flares more damaging than rotational spot modulation?

The present sample does not provide strong evidence that flares alone are more damaging than rotational spot modulation. Transit detectability appears to depend on the overall stellar variability environment rather than a single activity parameter. A larger sample would be required for a more definitive conclusion.

4.4 Are shallow transits disproportionately lost?

No strong evidence was found that shallow transits were disproportionately lost. The transit-depth distributions of Classes 2 and 3 are very similar, suggesting that stellar activity plays a larger role in determining detectability than planetary size.

4.5 Which model is most robust and why?

Random Forest was the most robust classifier, achieving perfect performance on the test dataset. The feature-based machine-learning models successfully exploited statistical properties of the light curves, allowing them to distinguish planetary transits from stellar variability with very high accuracy.

5 Conclusion

In this work, a simulated PLATO-like dataset was used to investigate the impact of stellar activity on exoplanet transit detection. Three machine-learning models—Random Forest, XGBoost, and a one-dimensional Convolutional Neural Network—were developed and evaluated.

All machine-learning models achieved excellent classification performance. Random Forest produced the best overall results, followed closely by XGBoost and CNN. Feature-importance analyses showed that planetary and variability-related parameters contain strong discriminatory information for exoplanet detection.

A classical Transit Least Squares baseline was also implemented. TLS achieved an overall recovery fraction of 53.3% on representative active-star light curves, significantly lower than the performance achieved by the machine-learning models.

The results demonstrate that machine-learning techniques provide a powerful framework for exoplanet detection under realistic stellar-activity conditions and offer substantial advantages over traditional transit-search methods.